DS8001 Project Proposal: Permutation-Based Approximate kNN Search for Mixed-Type Hospital Encounters

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1. Problem and question. Exact k-nearest-neighbor (kNN) search computes a distance to all N indexed objects. We study, as a purely algorithmic question (no clinical claim), how permutation-based approximate indexes trade recall for fewer distance computations under a mixed-type distance. **Principal question:** at matched recall@10, does the truncated MI-File (Amato and Savino, 2008) need fewer distance computations per query than the truncated permutation scan (TPS; Chávez et al., 2008), and how does this change with N? **Secondary:** how do the pivot count n, permutation length L and re-ranking budget c affect recall@10 and cost?

2. Methods and scope. Essential (implemented by us): (i) **brute-force exact kNN** (ground truth; ties by encounter id); (ii) **TPS**: each object stores the ranks of its L nearest pivots (rank L+1 otherwise); a query scores all objects by Spearman footrule over the n pivots, keeps the best c and re-ranks them with the true distance; (iii) **MI-File**: per-pivot posting lists of (object, rank) with the same score and re-ranking, touching only objects that share a retained pivot; random pivots. **Optional extensions (if time permits):** farthest-first pivots; scikit-learn **BallTree** as a library baseline (not counted as our implementation); extra grid values. Complexity: build O(Nn) distances, storage O(NL), query n pivot distances plus c re-ranking distances.

Distance. Six feature groups G (age, utilization, admission/discharge, ICD-9 diagnosis chapters, labs, medications) with equal weight 1/6. For objects x, y:

$$d(x,y) = (1/6) \sum_{g} (1/|g|) \sum_{f \in g} d_f(x_f, y_f) \in [0,1],$$

so each group is normalized by its feature count. Numeric/ordinal: $d_f = |x_f - y_f| / (\max_f - \min_f)$, range fit on the index and queries clipped to it. Nominal: $d_f = 1[x_f \neq y_f]$. **Missing values** are never omitted pairwise: diagnoses, specialty, payer and labs get an explicit UNKNOWN/NOT_MEASURED category (so two unknowns match, unknown vs. known mismatches); numerics use the index median. This is a weighted L1 metric, checked on sampled triples.

3. Data. Diabetes 130-US Hospitals (1999–2008) (Strack et al., 2014; UCI): 101,766 encounters, 71,518 patients. We exclude IDs, `readmitted`, race, gender and `weight`. Patients are split 90/5/5 into index/validation/test (fixed seed), so no patient is in both index and queries; scales and medians are fit on the index only.

4. Experiments, definitions and decision rules (fixed before running). Nested N = 10k, 50k, full index; 300 validation and 300 test queries; $n \in \{32, 64\}$, $L \in \{8, 16, 32\}$, $c/N \in \{0.5\%, 1\%, 5\%\}$; 3 seeds. **Query cost** = true-distance evaluations per query = n + c (machine-independent); postings read, build distances, index size, median latency and peak memory are secondary. **Operating point:** on validation, take each method's Pareto frontier (recall@10 vs. query cost) and select the lowest-cost setting with mean recall@10 $\geq \tau$ for each $\tau \in \{0.80, 0.90\}$; if a method cannot reach τ, that τ is reported as not reached for it (its highest-recall setting is still reported); ties go to smaller index size. Settings are then frozen and run once on test. **Principal comparison** (MI-File vs. TPS), at each N: (a) matched recall: cost at the selected points; (b) matched budget: recall@10 at equal n, L, c. **"Beats"** = lower mean query cost at the same τ (or higher recall at equal c), with the 95% patient-level bootstrap CI of the paired per-query difference excluding 0; otherwise we report a tie. We also report MI-File zero-overlap misses.

5. Validation and plan. Python 3.11+ (NumPy, pandas; scikit-learn only for the optional baseline), fixed seeds, pytest tests: metric axioms, TPS and MI-File equal brute force when c = N, no patient in two partitions. Already implemented: loader, patient-grouped split, encoder, brute-force oracle, metric tests. Plan: pivots and TPS (mid-Nov), MI-File (late Nov), experiments, analysis and report (by Dec 7); optional extensions only after the essential comparison is complete. **Threat:** one dataset and one distance; recall is w.r.t. our metric, not clinical similarity.

References

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